1 Preliminary notions

1.1 Probability theory

The foundation of modern probability theory is based on measure theory.

Definition 1.1 (σ-algebra)**.** A **σ-algebra** $\mathcal{F}$ on a set Ω is a collection of subsets of Ω that

- $\Omega \in \mathcal{F}$
- if $A \in \mathcal{F}$ then $X \setminus A \in \mathcal{F}$
- if $A_1, A_2, \ldots \in \mathcal{F}$ then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$

Definition 1.2 (Measure)**.** Let Ω be a set and $\mathcal{F}$ a σ-algebra on Ω. A function $\mu : \mathcal{F} \to [0, \infty]$ is **measure** if satisfies

1. $\mu(\emptyset) = 0$
2. $\mu(A) \geq 0$ for all $A \in \mathcal{F}$
3. if $\{A_i\}_{i=1}^{\infty} \in \mathcal{F}$ and $A_i \cap A_j = \emptyset$ for all $i \neq j$ then

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu\left(A_i\right)$$

If μ is a measure on $\mathcal{F}$ such that $\mu(\Omega) = 1$, then μ is called a **probability measure**.

Definition 1.3 (Probability space)**.** A probability space is a triple $(\Omega, \mathcal{F}, P)$ where

- Ω is the set of all possible outcomes
- $\mathcal{F}$ is a σ-algebra on Ω
- P is a probability measure.

For a more details in probability theory and stochastic processes, see [3] or [4].

1.2 Stochastic processes

Definition 1.4 (Stochastic processes)**.** A stochastic process is an indexed family of random variables

$$\{X(t, \omega)\}_{t \in T},$$

on a probability space $(\Omega, \mathcal{F}, P)$ assuming values in $\mathbb{R}^n$.

If the set T is countable then the stochastic process is said to be *discrete*, if T is some interval of $\mathbb{R}$ the is *continuous*.

An intuitive way to understand stochastic processes is to consider what is $X(t, \omega)$ when t or ω is fixed. If we fix $t \in T$ then the map

$$\omega \to X_t(\omega),$$

for $\omega \in \Omega$ is a random variable. If $\omega \in \Omega$ is fixed then the map

$$t \to X_\omega(t),$$

for $t \in T$ is a deterministic function, and is called *path* of $X(t, \omega)$.

Definition 1.5 (Filtration)**.** A **filtration** is a family of σ-algebras $\{\mathcal{F}_t\}_{t\in T}$ indexed by T, such that $\mathcal{F}_s \subseteq \mathcal{F}_t$ for all $s \leq t$, where $s, t \in T$.

The idea is that a filtration $\mathcal{F}_t$ represents the information available form the past of an stochastic process up to time t. Thus we define a **filtered probability space** as a probability space with a filtration $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\in T}, P)$.

Definition 1.6 (Adapted process)**.** Let X_t be a stochastic process on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\in T}, P)$. The process is **adapted** to the filtration $\{\mathcal{F}_t\}_{t\in T}$ if, for every $t \in T$, the random variable X_t is F_t-measurable.

The idea of an adapted process is key when modeling using stochastic processes, as it ensures that the process is at time t only depends on the information available up to that time, but not on future information.

1.2.1 Martingales

A martingale is a fundamental concept in the theory of stochastic processes, formalizes the idea of a "fair game", where given all the information up to now, the best prediction for the future is simply the present value.

Definition 1.7 (Martingale)**.** A stochastic process X_t where $t \in T$ is called a **martingale** with respect to a filtration F_t if it satisfies the following conditions:

1. $E[|X_t|] < \infty$, for each $t \in T$
2. X_t is measurable with respect to $\mathcal{F}_t$ for each $t \in T$
3. If X_t is discrete
$$E[X_{t+1}|F_t] = X_t \quad \forall t \in \mathbb{N},$$
if X_t is continuous
$$E[X_t|F_s] = X_s \quad \forall s \leq t.$$

1.2.2 Brownian Motion

Brownian motion (also called Wiener process) is the fundamental stochastic process underlying continuous-time financial modeling. Formally, a Brownian motion $\{W_t\}_{(t\geq 0)}$ is defined by the following properties:

1. $W_0 = 0$
2. The map $t \to W_t(\omega)$ is almost surely continuous
3. W_t has independent increments, i.e. For $t_1 < t_2 < t_3 < t_4$, $W_{t_2} - W_{t_2}$ and $W_{t_4} - W_{t_4}$ are independent random variables
4. W_t has gaussian increments, i.e. $W_t - W_s \sim \mathcal{N}(0, t-s)$ for $0 \leq s \leq t$

where $\mathcal{N}(\mu, \sigma)$ is the normal distribution with mean μ and variance σ.

The Brownian motion has some key features that will need afterwards, for example it can be shown that the paths $t \to W_t(\omega)$ although are almost surely continuous, they are nowhere differentiable.

Also the it can be shown that the Wiener process is an almost surely continuous martingale, and its quadratic variation is $\{W_t, W_t\} = t$. This two properties along $W_0 = 0$ form an equivalent definition of the Wiener process, called the Levy's characterization.

In [5] there are profs of this properties and the equivalence between the tow definitions, also there is a prof of the existence of the Brownian motion.

2 Stochastic calculus

We would model the dynamics of the value of a financial asset, an ordinary differential equation of the form

$$\frac{dS_t}{dt} = f(t, S_t),$$

would not be sufficient to capture the randomness present in the financial data.

Stochastic differential equations (SDEs) are similar to ODEs, but include a stochastic term that allows for modeling the random fluctuations,

$$\frac{dS_t}{dt} = f(t, S_t) + \sigma(t, S_t)W_t, \tag{1}$$

where the term W_t is a stochastic process. Notice that (1) can be written in a integration notation as

$$S_t = S_0 \int_0^t f(t, S_t)dt + \int_0^t \sigma(t, S_t)dW_t, \tag{2}$$

thus a key part of SDEs is defining the last term which is called stochastic integral.

The most well known process that has a well defined stochastic integral is the Wiener process or Brownian motion, this stochastic integral is known as the Ito integral and will allow us to have the machinery to study SDEs and thus model the dynamics of financial assets.

2.1 Ito's integral

Definition 2.1. Let $\mathcal{V} = \mathcal{V}(S, T)$ the class of functions

$$f(t, \omega) : [0, \infty) \to \mathbb{R},$$

such that

- (t, ω) is $\mathcal{B} \times \mathcal{F}$-measurable, where $\mathcal{B}$ is the Borel σ-algebra on $[0, \infty)$ and $\mathcal{F}$ is the σ-algebra on Ω,
- $f(t, \omega)$ is $\mathcal{F}_t$ adapted.
- $\mathbb{E}\left[\int_S^T f(t, \omega)^2 dt\right] < \infty$.

We will start by defining the Ito intregral for *elementary functions*, then this can be extended to functions in $\mathcal{V}$.

Definition 2.2. A function ϕ is called *elementary* if it has de form

$$\phi(t, \omega) = \sum_j e_j(\omega)\mathbb{1}_{[t_j, t_{j+1})}(t).$$

For elementary functions we define the Ito integral as

$$\int_S^T \phi(t, \omega)dW_y(\omega) = \sum_j e_j(\omega)\left[B_{t_{j+1}} - B_{t_j}\right](\omega).$$

Lemma 2.1 (Ito isometry)**.** Let $\phi(t, \omega)$ be a bounded elementary function, then

$$\mathbb{E}\left[\left(\int_S^T \phi(t, \omega)dB_t(\omega)\right)^2\right] = \mathbb{E}\left[\int_S^T \phi(t, \omega)^2 dt\right].$$

Using the Ito isometry and the definition of the Ito integral for elementary functions, the Ito integral can be extended to the class of functions $\mathcal{V}$, using that the functions in $\mathcal{V}$ can be expressed as the limit of sequences of simpler functions. It is worth mention the the Ito isometry still holds for any function in $\mathcal{V}$.

The result of the integral is a stochastic process, which is a continuous martingale. This is similar to regular integration, where the result of integrating a function is a new function, in this case the result of integrating a process "over" a stochastic process is a new stochastic process, plus the resulting process has the nice property of being a martingale.

The class $\mathcal{V}$ is rather limited and does not even contain all continuous functions, see [7]. Having to restrict to this class would be rather inconvenient when modeling using SDEs, luckily the Ito integral can be extended to broader class by relaxing the last condition of the definition 2.1

$$P\left(\int_S^T f(t,\omega)^2 dt < \infty\right) = 1,$$

will call this class of functions $\mathcal{W}$. Notice that the class $\mathcal{W}$ does contain all continuous functions.

2.2 Ito's formula

Having defined the Ito integral, would be interesting to have a way to easily compute stochastic integrals without having to compute the limit of elementary functions, that is the purpose of the Ito's lemma or Ito's formula. The Ito's formula is a stochastic version of the chain rule for derivatives.

Definition 2.3 (Ito process)**.** Let B_t be a Brownian motion on $(\Omega, \mathcal{F}, P)$. An Ito process is a stochastic process X_t on $(\Omega, \mathcal{F}, P)$ of the form

$$X_t = X_0 + \int_0^t \mu(s, X_s) ds + \int_0^t \sigma(s, X_s) dB_s, \tag{3}$$

where μ and σ are adapted, measurable processes, that satisfy the integrability conditions

$$P\left(\int_0^T |\mu(t,\omega)|\, dt < \infty\right) = 1,$$

and

$$P\left(\int_0^T \sigma(t,\omega)^2 dt < \infty\right) = 1.$$

The equation (3) can be also written in differential form as

$$dX_t = \mu ds + \sigma dB_s.$$

Theorem 2.2 (1-dimensional Ito's formula)**.** Let X_t be an Ito process given by

$$dX_t = \mu ds + \sigma dB_s.$$

Let $g(t,x) \in C^2([0,\infty) \times \mathbb{R})$, then

$$Y_t = g(t, X_t),$$

is an Ito process given by

$$dY_t = \frac{\partial g}{\partial t}(t, X_t) dt + \frac{\partial g}{\partial x}(t, X_t) dX_t + \frac{1}{2}\frac{\partial^2 g}{\partial x^2}(t, X_t)(dX_t)^2, \tag{4}$$

where $(dX_t)^2$ is computed according to the rules

$$(dB_t)^2 = dt, \qquad (dt)^2 = 0, \qquad dB_t \cdot dt = 0.$$

For the proof of the Ito's formula we refer to [6]. The formula can be extended to multiple dimensions.

Definition 2.4 (n-dimensional Ito process)**.** Let $B_t = (B_1(t), \ldots, B_m(t))$ be a m-dimensional Brownian motion, let μ_i and σ_{ij} be adapted, measurable processes such that

$$P\left(\int_0^T |\mu_i(t,\omega)|\, dt < \infty\right) = 1, \qquad P\left(\int_0^T \sigma_{ij}(t,\omega)^2 dt < \infty\right) = 1,$$

then we define the n-dimensional Ito process $X_t = (X_1(t), \ldots, X_n(t))$ as

$$\begin{cases} dX_1(t) = \mu_1 dt + \sigma_{11} dB_1(t) + \cdots + \sigma_{1m} dB_m(t), \\ \vdots \\ dX_n(t) = \mu_n dt + \sigma_{n1} dB_1(t) + \cdots + \sigma_{nm} dB_m(t), \end{cases}$$

or in matrix notation

$$dX_t = \mu ds + \sigma dB_s.$$

where $\mu = (\mu_1, \ldots, \mu_n)$ and $\sigma = (\sigma_{ij})$ are n-dimensional vectors and $m \times n$ matrices respectively.

Theorem 2.3 (General Ito's formula)**.** Let X_t be a n-dimensional Ito process given by

$$dX_t = \mu ds + \sigma dB_s.$$

Let $g(t,x) = (g_1(t,x), \ldots, g_p(t,x)) \in C^2([0,\infty) \times \mathbb{R}^n)$, then

$$Y_t = g(t, X_t),$$

is a p-dimensional Ito process whose components are given by

$$dY_k(t) = \frac{\partial g_k}{\partial t}(t, X_t)dt + \sum_i \frac{\partial g_k}{\partial x_i}(t, X_t) dX_i(t) + \frac{1}{2}\sum_{ij} \frac{\partial^2 g_k}{\partial x_i \partial x_j}(t, X_t) dX_i(t) dX_j(t), \tag{5}$$

where $dX_i(t)dX_j(t)$ is computed according to the rules

$$(dB_i(t))^2 = dt, \qquad dB_i(t)dB_j(t) = 0.$$

2.3 Stochastic differential equations

3 Black-Scholes

Black-Scholes is a mathematical model for pricing options. Although many other models have been developed since, the Black-Scholes model remains as the most important model in option pricing, as in many cases options in markets are quoted using the Black-Scholes formula.

A derivative is a financial instrument whose payments are derived from the price of an other asset, called the underlying asset. Derivatives are widely used in finance, whether to hedge financial risk, this is to protect against adverse price movements, or to bet on specific views, especially when it's difficult to take a position in the underlying asset. So being able to price them and justify the price is key for the market to function.

3.1 Replication and arbitrage

One of the most important concepts in modern finance is the so called *law of one price*. The idea is that if two financial assets have exactly the same payoffs in all possible scenarios, then they should have the same price.

This is a linked to the concept of *arbitrage*, informally we say that arbitrage is the opportunity to make a riskless profit. For example, if two assets that have exactly the same payoffs in all possible scenarios, but don't follow the law of one price, one can buy the cheaper one and sell the more expensive one, gaining and instant profit and compensating the payoffs of the sold asset using the payoffs of bought one, which match exactly.

We will work under the assumption that we are in operating in a market with no arbitrage, this assumption is technically known as *The First Fundamental Theorem of Asset Pricing*, and a key part of the *Efficient Market Hypothesis*. Under this assumption, we can price any derivative by creating a replicating portfolio, this is a set of assets that has exactly the same payoffs as the derivative in all possible scenarios.

There are two main ways of creating a replicating portfolio:

- **Static replication** refers to an replicating portfolio that is created at the beginning and held as is until maturity.

- **Dynamic replication** refers to a replicating portfolio that requires a continuous change of its composition over time.

For a further discussion on replication, and its applications in finance, we recommend the book [2].

Will show two examples of static replication from [7] that will be useful to understand the Black-Scholes model.

3.1.1 Example: Forward contract

Assume that there is an agent that borrows and lends money at a continuously compounded rate r, known as the **risk-free rate**. This means that if we borrow 1$ at time $t = 0$, we will have to pay $e^{rT}$$ at time $t = T$.

A forward contract gives the holder the obligation to buy an asset at a fixed price K at a fixed date T.

To price this contract imagine we are at time t and the current price of an asset is S, we want to buy a forward contract with maturity T and fixed price K.

We have two options, we can buy the contract at time t for a price of F and at time T pay K to get the asset. Alternatively we can buy the asset for S and borrow $e^{-r(T-t)}K$

at time t, so at time T we will pay $K = e^{r(T-t)}e^{-r(T-t)}K$ while having the asset. Both options have the same cashflow at time T, pay K and own the asset, so either we have

$$F = S - e^{-r(T-t)}K,$$

or there is an arbitrage opportunity.

We have found the price of the forward contract by replicating its cashflows, using the underlying assed and the risk-free rate, plus arguing that there should not be an arbitrage opportunity.

3.1.2 Example: Put-Call parity

Definition 3.1 (European call option)**.** An **European call option** is a financial contract that gives the holder the right, but not the obligation, to **buy** an asset at a fixed price (the strike price) at a fixed date (the maturity date).

The payoff of an European call option at maturity T is given by:

$$C_T = \max(S_T - K, 0), \tag{6}$$

where S_T is the price of the underlying asset at maturity and K is the strike price.

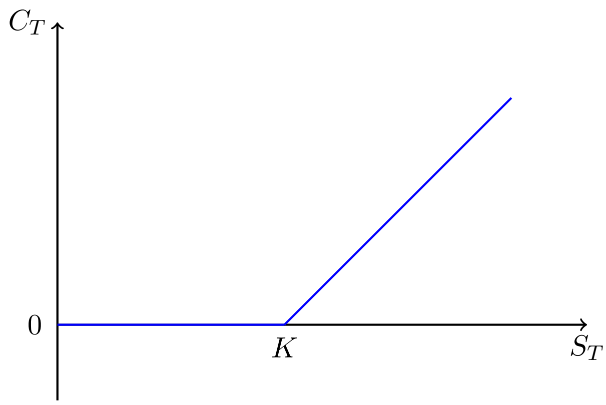

Figure 1: Payoff of a European call option

Definition 3.2 (European put option)**.** An **European put option** is a financial contract that gives the holder the right, but not the obligation, to **sell** an asset at a fixed price (the strike price) at a fixed date (the maturity date).

The payoff of an European put option at maturity T is given by:

$$P_T = \max(K - S_T, 0), \tag{7}$$

where S_T is the price of the underlying asset at maturity and K is the strike price.

Now consider a portfolio where we buy a call option and sell a put option with same strike K and maturity T, the payoff of such portfolio at time T will be exactly $S_T - K$. So we are essentilly reciving the asset and paying K at time T, as we have seen this is the payoff of a forward contract, which means that at time t the price of our portfolio must be equal to the price of a forward of the asset,

$$C_t - P_t = F_t = S_t - e^{-r(T-t)}K, \tag{8}$$

this formula is called *put-call parity* and illustrates the relationship between the price of a call and a put option.

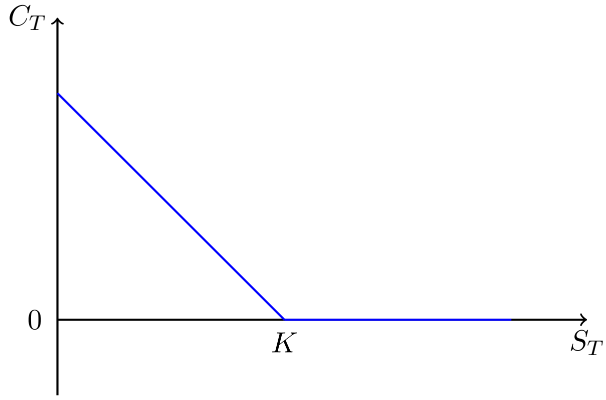

Figure 2: Payoff of a European put option

3.2 The Black-Scholes model

The Black-Scholes model is a mathematical model for pricing options, that is based on replicating the payoffs of the option dynamically using the underlying asset and a risk-free rate bond.

Like before we assume that the market is arbitrage-free, and that there is a constant risk-free rate r. As the model is based on dynamic replication, we also need to assume that we can rebalance our replicating portfolio continuously, i.e. we can buy and sell any amount of the underlying asset and the risk-free rate bond instantaneously at any time.

Let S_t be the price of the underlying asset at time t and β_t be the price of the risk-free rate bond at time t. The dynamics of this two assets are given by the following stochastic differential equations (SDEs):

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \qquad d\beta_t = r\beta_t dt.$$

To construct the replicating portfolio. Let a_t be the amount of the underlying asset in our replicating portfolio and b_t be the amount of the risk-free rate bond in our replicating portfolio. Then the value of the portfolio at any time t is given by:

$$V_t = a_t S_t + b_t \beta_t.$$

We need this portfolio to have the same payoff as the option at time T, so

$$V_T = \max S_T - K, 0.$$

As the portfolio is dynamically replicating the option, the cashflow of the portfolio must match the ones of the option, these are the payoff as seen before, and the initial payment, so no cash must came in or out of the portfolio at any time between t and T. This means that the portfolio must be *self-financing*, so the rebalanced must come from selling the underlying asset and buying the bond, or vice versa. This is expressed mathematically as

$$dV_t = a_t dS_t + b_t d\beta_t, \tag{9}$$

which can be extended to

$$dV_t = \left(a_t \mu S_t + b_t r \beta_t\right) dt + a_t \sigma S_t dW_t.$$

Now as the value of the replicanting portfolio V_t should be equal to the value of the option at any time t, we assume that this can be expressed by a function $f(t, S_t)$, depending only on the time t and the price of the underlying asset S_t. Thus applying Ito's formula we have that

$$dV_t = \left(\frac{\partial f}{\partial t} + \frac{\partial f}{\partial S}\mu S_t + \frac{1}{2}\frac{\partial^2 f}{\partial S^2}\sigma^2 S_t^2\right) dt + \frac{\partial f}{\partial S}\sigma S_t dW_t,$$

matching the two expressions for dV_t we have that

$$a_t = \frac{\partial f}{\partial S},$$

and

$$b_t = \frac{1}{r\beta_t}\left(\frac{\partial f}{\partial t} + \frac{1}{2}\frac{\partial^2 f}{\partial S^2}\sigma^2 S_t^2\right)$$

as V_t is the value of the replicating porfolio and the option at time t, we have that

$$f(t, S_t) = V_t = a_t S_t + b_t \beta_t,$$

using the values of a_t and b_t from above,

$$f(t, S_t) = \frac{\partial f}{\partial S} S_t + \frac{1}{r}\left(\frac{\partial f}{\partial t} + \frac{1}{2}\frac{\partial^2 f}{\partial S^2}\sigma^2 S_t^2\right),$$

that can be rearranged to

$$\frac{\partial f}{\partial t} = rf(t, S_t) - r\frac{\partial f}{\partial S} S_t - \frac{1}{2}\frac{\partial^2 f}{\partial S^2}\sigma^2 S_t^2, \tag{10}$$

with terminal condition

$$V_T = f(T, S_T) = \max(S_T - K, 0),$$

which is the **Black-Scholes PDE**.

4 Arbitrage theory

We can generalize the approach used to derive the Black-Scholes PDE for a more general products Thus in this section will try to give a general framework to derive equations to price a *contingent claim*.

Definition 4.1 (Market)**.** A market is a set of $n+1$ assets $S_0, S_1, \ldots, S_n$, where the price of S_0 is given by

$$\frac{dS_0(t)}{S_0} = r(S_0(t), t)dt,$$

and the prices of the other assets are given by

$$dS_i(t) = \mu(S_i(t), t)dt + \sigma_i(S_i(t), t)^{\intercal} dW(t),$$

where $W(t)$ is a k-dimensional Brownian motion.

Definition 4.2 (Portfolio)**.** A portfolio in the market $\{S_i(t)\}$ is an $n+1$ dimensional $\mathcal{F}_t^{(k)}$-adapted stochastic process

$$\theta(t) = (\theta_0(t), \theta_1(t), \ldots, \theta_n(t)),$$

an its value at time t is given by

$$V(t) = \sum_{i=0}^{n} \theta_i(t) S_i(t).$$

Note that as a portfolio $\theta(t)$ is an adapted process it doesn't use future information, so it is a generalization on what in reality would be to rebalance ones investments in the market at time t.

We can generalize the self-financing condition presented in (9) as,

$$V(t) = V(0) + \int_0^t \theta(u) dS(u), \tag{11}$$

in differential form

$$dV(t) = \theta(u) dS(t). \tag{12}$$

A contingent claim is a financial instrument whose payoff depends on the outcome of uncertain future events, for example an European option. As our market consist of assets that are descried as stochastic processes, an European contingent claim paying at time T is a function $f(S(T), T)$ depending on these assets, which is a random variable as $S(T)$ is a random vector of the values of the assets at time T.

4.1 PDE method

Consider an European contingent claim with payoff $f(S(T))$, similar to what we did in section 3.2, we want to construct a replicating portfolio $V(S(t), t)$ that has the same payoff as the contingent claim at time T. Thus using the Ito formula (5) we have

$$\begin{aligned} dV(S(t), t) &= \frac{\partial V}{\partial t} dt + \sum_{i=0}^{n} \frac{\partial V}{\partial S_i} dS_i(t) + \frac{1}{2} \sum_{i,j} \frac{\partial V^2}{\partial S_i \partial S_j} dS_i(t) dS_j(t) \\ &= \sum_{i=0}^{n} \frac{\partial V}{\partial S_i} dS_i(t) + \left(\frac{\partial V}{\partial t} + \frac{1}{2} \sum_{i,j} \frac{\partial V^2}{\partial S_i \partial S_j} \sigma_i(S(t), t)^{\intercal} \sigma_j(S(t), t) \right) dt, \end{aligned}$$

imposing the self-financing condition (12) we find that

$$\theta_i(t) = \frac{\partial V}{\partial S_i}, \quad \forall i = 0, \dots, n, \tag{13}$$

and

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sum_{i,j} \frac{\partial V^2}{\partial S_i \partial S_j} \sigma_i(S(t),t)^\intercal \sigma_j(S(t),t) = 0. \tag{14}$$

Also using $V(S(t),t) = \theta(t)S(t)$ in (13) we get

$$V(S(t),t) = \sum_{i=0}^{n} \frac{\partial V}{\partial S_i} S_i. \tag{15}$$

The equations (14) and (15) describe the PDE for the price of the contingent claim, using as boundary condition the payoff at time T

$$V(S(T),T) = f(S(T)).$$

As in Black-Scholes model, μ does not explicitly appear in the PDE, its only present implicitly in the prices $S(t)$ of the assets.

4.2 Martingale method

The martingale method is an alternative framework to the PDE method, that allows to price contingent claims in a more flexible way.

We will use [1] as a reference for this section, we start with a couple of definitions related with the concept of market and portfolio defined previously.

Definition 4.3 (Attainable process)**.** We say a price process $V(t)$ is *attainable* if there exists a self-financing portfolio $\theta(t)$ such that $V(t) = \theta(t)S(t)$.

The attainable process is a formalization of the dynamic replicating portfolio idea. An other restriction on the portfolios we are going to consider is the following,

Definition 4.4 (Admissible portfolio)**.** A portfolio $\theta(t)$ is called *admissible* if exists $\alpha > 0$ such that

$$\int_0^t \theta(t) dS(t) \geq -\alpha, \quad \forall t \in [0,T].$$

In real life we can interpret this condition as a limit on the amount the portfolio can lose or borrow, as bank, clearing house or counterparty will not allow such risks and would force to cancel the deal. Technically also allow to cancel out some other undesirable strategies that would prevent some key theorems to hold, see [7] sections 14.5 and 14.6 for more details on such strategies.

Now we will formalize the idea of arbitrage in a market.

Definition 4.5 (Arbitrage)**.** A self-financing portfolio $\theta(t)$ is an *arbitrage* if one of the following conditions hold for time t:

1. $\theta(0)S(0) < 0$ and $P(\theta(t)S(t) \geq 0) = 1$,
2. $\theta(0)S(0) = 0$ and $P(\theta(t)S(t) \geq 0) = 1$ and $P(\theta(t)S(t) > 0) > 0$,

Consider that the *risk-free* asset S_0 has a constant price of 1, i.e. the *risk-free* rate is $r(S(t), t) \equiv 0$. Under this assumption we introduce the *risk-neutral measure* $\mathcal{Q}$ as,

Definition 4.6. A probability measure $\mathcal{Q}$ on $\mathcal{F}_T$ is called a *risk-neutral measure* (also known as *equivalent martingale measure*) if

- $\mathcal{Q}$ is equivalent to P on $\mathcal{F}_T$.
- All price process in the market $S_0, S_1, \ldots, S_n$ are martingales under $\mathcal{Q}$ on the time interval $[0, T]$.

We refer to P as the "real" probability measure and reflects the true probability of the price processes, whereas $\mathcal{Q}$ is a probability measure under which the expected price of any asset is equal to the current one regardless of their risk (volatility).

5 Pricing - Solving the Black-Scholes PDE

We have found the Black-Scholes PDE for European calls and we have seen the put-call parity, so assuming that we know the volatility σ and the risk free rate r, we can know the fair price of an European option.

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